

src/app/commitChangeAnalysis.ts

Kind: added · Baseline: f7c2bfd81af2 · Target: NovaDiff · Generated: 2026-05-21T10:53:08.972Z

Overview

A new file `src/app/commitChangeAnalysis.ts` (lines 1–209) introduces browser-safe utilities for classifying commit changes. It replaces external graph-core types with local interfaces and removes `node:fs` dependencies.

Key changes

- **Interfaces** (`ChangeAnalysis` , `UpdateDecision`) defined at lines 4–19.
- `topDirectory` (lines 21–27) extracts the first path segment or returns `null` .
- `detectDirectoryChanges` (lines 29–49) flags structural changes when new or deleted files add previously unknown top-level directories.
- `summarizeChanges` (lines 52–63) builds a concise string of new, deleted, and modified file counts.
- `classifyUpdate` (lines 67–129) implements decision logic:
 - `SKIP` if no structural changes.
 - `FULL_UPDATE` when structural changes exceed 30 files or 50 % of the graph.
 - `ARCHITECTURE_UPDATE` if directory changes or >10 structural files.
 - `PARTIAL_UPDATE` otherwise.
- `buildChangeAnalysisFromRows` (lines 132–152) converts `FileChange` rows into a `ChangeAnalysis` .
- `classifyCompareRows` (lines 155–163) wraps the above for direct use.
- `districtBreakdown` (lines 165–185) aggregates changes by top-level directory and sorts by count.
- `graphSymbolsContext` (lines 188–209) lists affected functions/classes from a graph, limited to 12 entries.

Impact

- New public API: modules can import `classifyCompareRows` or `districtBreakdown` without pulling in Node-specific code.
- No existing code is altered; the file is entirely new.
- `detectDirectoryChanges` runs in $O(n)$ over file lists, suitable for typical commit sizes.
- Human-readable summaries (`summarizeChanges` , `graphSymbolsContext`) aid UI or log output.

Risks & follow-ups

- **Threshold correctness:** the >30 and >50 % rules are hard-coded; verify they match project policy.
- **Directory detection:** `topDirectory` may mis-handle paths with leading slashes or empty segments.

- **Graph node filtering:** `graphSymbolsContext` may miss symbols when `changedPaths` contains nested files.
- **Test coverage:** add unit tests for all decision branches and for `districtBreakdown` aggregation to guard against regressions.